BME-DALEEL at Daleel 2026: Joint Modeling, Structured Decoding, and Domain-Routed Ensembles for Arabic Argument Units

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Abstract

We present BME-DALEEL, our system for both Daleel 2026 Arabic argumentative discourse mining tasks. A joint Transformer predicts paragraph-level label presence and token-level BIO states; constrained decoding and domain-routed ensembles map spans to original character coordinates. Repeated out-of-fold comparisons gate domain specialists, while routing selected decoded outputs preserves other-genre rows exactly. Using organizer annotations and public pretrained models, our Closed-track system scores 0.7117 macro-F1 for paragraph classification and 0.7547 modified F1 for span detection.

1 Introduction

Argument mining identifies units that express claims and evidence (Lawrence and Reed, 2019). Arabic remains comparatively under-resourced at this level; Daleel 2026 addresses this gap with Arabic newspaper editorials and competitive debates (Nabhani et al., 2026).

Our main challenge is to combine diverse encoders given 217 development paragraphs. We therefore admit sources, weights, and decoder biases only through repeated out-of-fold comparisons. The resulting system contributes (i) joint paragraph and span prediction with legal BIO transitions and deterministic reconstruction on original character offsets, (ii) validation-gated domain specialists, and (iii) post-decode routing with an exact branch non-interference invariant. Code and frozen configurations are public at `github.com/Mo7aisen/BME-Daleel-2026`.

2 Background: Task and Data

Daleel defines six discourse-unit labels: assumption (AS), anecdote (AN), statistics (ST), testimony (TE), common ground (CO), and other (OT). TASK 1 returns the labels present in a paragraph; TASK 2 returns labeled half-open character intervals. A synthetic quoted authority followed by an unsupported claim, for example, yields $\{\mathrm{TE}, \mathrm{AS}\}$ and spans such as $(\mathrm{TE}, [12, 41))$ and $(\mathrm{AS}, [45, 88))$.

We use only organizer train and development labels. Train contains 255 editorial and 357 debate paragraphs (2,975 spans), development 88 and 129 (961 spans), and hidden-label test input 87 and 126. Text is preserved without normalization, translation, or external segmentation. TASK 1 uses six-label macro-F1; TASK 2 uses released modified F1, with fractional precision and recall credit for same-label character overlap. This paper reports both tasks in the Closed resource setting.

3 Related Work

Arabic argument-mining resources include projected multilingual news annotations (Sliwa et al., 2018), the Munazarat debate corpus (Khader et al., 2024), a higher-level hybrid debate scheme (Al-Sharafi et al., 2025), and cross-lingual transfer from non-Arabic labels (Nabhani and Al Khatib, 2025). Daleel instead provides native, function-labeled character spans in two Arabic genres; we neither translate nor import labeled examples.

Joint global/local prediction pairs intent with slots (Zhang et al., 2019), sentence classes with definition spans (Pouran Ben Veyseh et al., 2020), and argument sequence labels with roles or relations (Stab and Gurevych, 2017; Eger et al., 2017). We add constrained decoding, validated encoder admission, and post-decode routing. The pool covers XLM-R (Conneau et al., 2020), MARBERT (Abdul-Mageed et al., 2021), CAMeLBERT (Inoue et al., 2021), mDeBERTaV3 (He et al., 2023), and JHU English–Arabic (Yarmohammadi et al., 2021).

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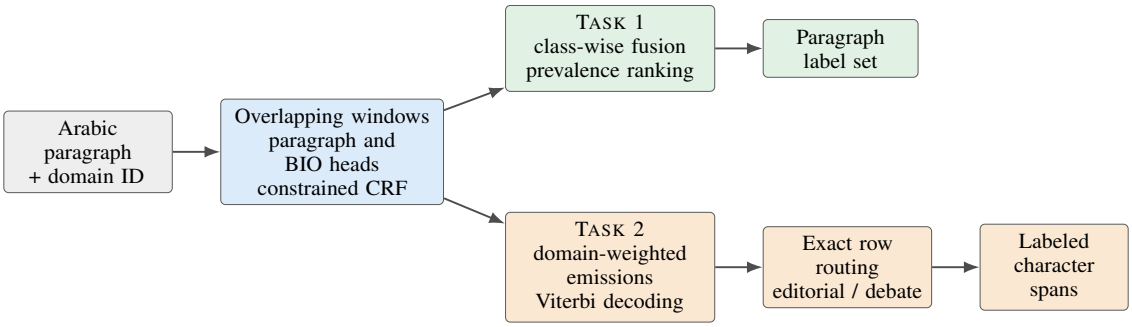


Figure 1: System overview. Training is joint; fusion, count selection, and final routing are domain-specific.

4 System Overview

Figure 1 summarizes the shared model and task-specific decision paths.

4.1 Joint paragraph and span model

Each fast tokenizer produces windows of at most 512 subwords with stride 96. Character offsets are retained, and token probabilities across overlapping windows are averaged with a symmetric linear distance-from-edge ramp. Gold spans map to O plus B/I states for the six labels. Where annotations overlap, BIO labels use fixed priority ST>CO>TE>AN>OT>AS.

A learned two-valued domain embedding is added to each Transformer representation h_i (Vaswani et al., 2017) and layer-normalized; three dropout views are averaged. The token head emits 13 BIO scores, while the paragraph head concatenates the first-token representation and masked mean pooling to predict six sigmoid probabilities. Initial loss is $\mathcal{L}_{\text{CE}} + \lambda_p \mathcal{L}_{\text{BCE}}$, with 0.02 token label smoothing and frequency weights. Selected models continue with a linear-chain CRF (Lafferty et al., 2001). Illegal starts and I-label transitions receive -10^4; structured loss is length-normalized CRF negative log likelihood plus $0.5\mathcal{L}_{\text{CE}}$.

4.2 Task-specific fusion and routing

TASK 1 combines five probability sources: MARBERTv2 and XLM-R paragraph/token blends, a character/word TF–IDF logistic model, CRF-decoded presence, and maximum CRF marginal posterior. CRF-D/P derive from the XLM-R/CAMeLBERT/mDeBERTa CRF ensemble. Per-label convex weights and domain thresholds are fitted on repeated folds. Rows are ranked within domain and label, and the top k are selected from train+development prevalence.

Table 1 reports selected weights: CRF marginals receive high weight for rare ST, TE, and CO, while the lexical model contributes to AS and OT. For domain d with n_d test rows and label ℓ, we return the $k_{d\ell} = \text{round}(n_d c_{d\ell}/N_d)$ highest probabilities, where $c_{d\ell}/N_d$ is the train+development occurrence rate. Thus prediction counts are set without hidden test labels.

Label	MAR	XLM	TF–IDF	CRF-D	CRF-P
AS	.1	.0	.4	.0	.5
AN	.2	.3	.0	.2	.3
ST	.1	.0	.0	.0	.9
TE	.3	.0	.0	.0	.7
CO	.0	.3	.0	.0	.7
OT	.0	.5	.4	.1	.0

Table 1: Final TASK 1 convex source weights. CRF-D is decoded presence; CRF-P is the marginal posterior.

For TASK 2, CAMeLBERT and mDeBERTa token distributions are projected to the XLM-R grid by character-overlap weighting, with XLM-R supplying CRF transitions. The base three-source mixture, ordered as XLM-R, CAMeLBERT, and mDeBERTa, uses weights $(0.8, 0.1, 0.1)$. Editorials add an XLM-R continuation directly as a fourth source with weights $(0.6, 0.1, 0.1, 0.2)$. Debates instead blend the base bundle at 0.8 with a JHU English–Arabic source at 0.2, giving effective four-source weights $(0.64, 0.08, 0.08, 0.20)$. After Viterbi decoding, editorial rows are copied from the frozen editorial branch and debate rows from the JHU-fused branch. No decode, presence filter, or span merge follows routing.

For target token interval I_t and source interval I_s, let $o_{ts} = |I_t \cap I_s|$. Source probabilities are projected as

$$\widetilde{p}_t = \frac{\sum_s o_{ts} p_s}{\sum_s o_{ts}}, \tag{1}$$

with the target distribution retained when the denominator is zero. Projection therefore supports differing tokenizer grids while preserving the primary

grid. Decoded BIO runs are mapped back through stored half-open offsets, trimmed only at whitespace and clause punctuation, and deduplicated in character order. "Character-exact reconstruction" means that each output is a direct slice of the original input; it does not claim exact agreement with gold boundaries.

4.3 Post-decode routing and non-interference

Let $\hat{y}_i^E$ and $\hat{y}_i^D$ be the already decoded editorial and debate branch outputs. Final routing is the deterministic operation

$$\hat{y}_i = \begin{cases} \hat{y}_i^E, & d_i = \text{editorial}, \\ \hat{y}_i^D, & d_i = \text{debate}. \end{cases} \tag{2}$$

This is not another soft ensemble: the chosen structured output is copied exactly. Consequently, admitting the debate specialist cannot change an editorial row, and vice versa. Routing thus preserves each selected branch's domain rows exactly and occurs after, rather than before, Viterbi decoding.

5 Experimental Setup

Table 2 identifies every final neural source. The full Hugging Face IDs are *FacebookAI/xlm-roberta-large*, *UBC-NLP/MARBERTv2*, *CAMeL-Lab/bert-base-arabic-camelbert-mix*, *microsoft/mdeberta-v3-base*, and *jhu-clsp/roberta-large-eng-ara-128k*. Full revision hashes and stage-specific settings are in the public configuration. The paragraph-loss coefficient λ_p is 0.7 for MARBERT, 0.6 for other initial fits, and 0.3 during CRF continuation.

All final neural models use AdamW (Loshchilov and Hutter, 2019), cosine decay, 0.05 warmup, weight decay 0.01, gradient clipping at 1.0, BF16, and deterministic seeds. Software is Python 3.11, PyTorch 2.4.1, Transformers 4.46.3, and scikit-learn 1.5.x. Training used one NVIDIA A100 40GB GPU on HUN-REN Cloud (Héder et al., 2022).

Three repeated balanced three-fold splits refit TASK 1 fusion and count selection. TASK 2 decoder fits use the same structure; the selected JHU weight is evaluated in 20 newly seeded three-fold repeats. No neural checkpoint is refitted inside these meta-level folds. After decisions are frozen, final checkpoints use train+development. Canonical JSONL, fixed ZIP timestamps, SHA-256 hashes, and fourteen tests verify all 213 test rows and a bit-for-bit archive rebuild. No external annotations, translation, pseudo-labels, or generated examples enter training.

6 Results and Analysis

Post-submission joint diagnostic. These organizer-development-set, pre-CRF XLM-R diagnostics share split and schedule, with the best development epoch selected separately per task ($n = 3$; mean±sample SD). For TASK 1, the joint model's tuned paragraph/token blend scores 0.7447 ± 0.0187 versus 0.6246 ± 0.0215 for the Task-1-only paragraph head; identical TASK 2 token decoding gives 0.6611 ± 0.0267 joint versus 0.6962 ± 0.0072 token-only. Paired differences are $+0.1201 \pm 0.0287$ and -0.0351 ± 0.0335. Because TASK 1 changes both loss and inference evidence, its delta describes a configuration rather than a causal joint-loss effect. Joint modeling is therefore not uniformly beneficial: under this development-selected setup, TASK 2 favors token-only training. This post-submission diagnostic did not alter official entries.

6.1 Out-of-fold comparisons

Table 3 reports repeated out-of-fold comparisons. CRF evidence raises TASK 1 macro-F1 within this procedure. For TASK 2, the frozen base mean is 0.7550 ± 0.0061 modified F1. The editorial continuation has mean paired changes of $+0.00348$ overall (17/20 positive) and $+0.01296$ editorial (18/20 positive), with debate unchanged. The JHU source then has mean paired changes of $+0.00791$ overall (19/20 positive) and $+0.01185$ debate (20/20 positive), with editorial unchanged. A proposed TASK 1 editorial replacement was rejected over 100 repeats (mean delta -0.00565, paired-bootstrap 95% CI $[-0.00730, -0.00398]$).

6.2 Official results

Table 4 gives the CodaBench scores for our Closed-track entries. Each task is reported once.

6.3 Error analysis

We descriptively audited stable full-development predictions; these are not untouched estimates. TASK 1 recovered the exact label set for 129/217 paragraphs, with 20 missed OT labels and 17 excess predictions each for AS and AN. For TASK 2, each gold independently takes the prediction with largest positive character intersection; ties take the lexicographically earliest (start, end, label). Matching is not one-to-one: one prediction may serve

Source (revision)	Final role	Initial epochs	Later epochs	Batch	Seed	Encoder / head learning rate
MARBERTv2 (`fe88db9`)	TASK 1	10	–	16	42	$2e{-}5/8e{-}5$
XLM-R-large (`c23d21b`)	Both	10	1 + 2 CRF	18	43/61	$1e{-}5/5e{-}5$; CRF $3e{-}6/3e{-}5 \rightarrow 2e{-}6/2e{-}5$
CAMeLBERT-mix (`9be3527`)	Both	11	4 CRF	16	55/62	$2e{-}5/8e{-}5$; CRF $3e{-}6/3e{-}5$
mDeBERTa-v3-base (`a048466`)	Both	8	2 CRF	16–20	44/64	$2e{-}5/8e{-}5$; CRF $3e{-}6/3e{-}5$
XLM-R editorial	TASK 2	–	4 continuation	18	92	$1e{-}6/1e{-}5$
JHU English–Arabic (`c1a082e`)	TASK 2	10	1 + 2 CRF; 4 refit	18	93/94	$1e{-}5/5e{-}5$; final $1e{-}6/1e{-}5$

Table 2: Training schedules. Batch is effective batch size; revisions are Git commit prefixes.

System	Overall	Editorial	Debate
TASK 1 base (3 sources)	0.7052	0.6811	0.7157
+ decoded CRF presence	0.7085	0.6879	0.7097
+ CRF marginal posterior	**0.7189**	**0.6908**	**0.7225**
TASK 2 frozen ensemble	0.7543	0.6980	0.7762
+ JHU debate source (0.20)	**0.7622**	0.6980	**0.7880**

Table 3: Repeated out-of-fold F1: 3×3 for TASK 1 and 20 newly seeded three-fold repeats for the JHU comparison.

Task	Track	Overall	Editorial	Debate
TASK 1	Closed	0.7117	0.6239	0.7778
TASK 2	Closed	0.7547	0.6653	0.7941

Table 4: Closed-track test scores for the submitted system.

multiple gold spans. Of 961 gold spans, 153 are exact (15.9%), 567 have the same label but wrong boundaries (59.0%), 236 substitute a label (24.6%), and five have no predicted overlap (0.5%). Table 5 shows that AN→AS, OT→AS, and TE→AS are most frequent. Under this diagnostic, boundary-only mismatches are more frequent than label substitutions.

One paraphrased debate error expanded a 39-character anecdote into a 313-character AN span. Three of five nonoverlapping OT units were only 4–12 characters long, suggesting split/merge and short-unit modeling.

7 Limitations and Ethical Considerations

Only 47 ST and 66 CO spans occur in train+development, leaving rare-label estimates unstable. BIO cannot represent nested units; character projection can blur boundaries; prevalence ranking assumes limited shift. Full-refit checkpoints lack untouched development estimates, so evidence comes from pre-refit comparisons. Labels describe discourse function—not truth, intent, or credibility—and must not be used to judge speakers. We add no personal data, external annotation, generated examples, or hidden test references.

Gold \ predicted	AS	AN	TE	OT
AS	–	19	16	7
AN	51	–	16	1
ST	3	1	5	0
TE	47	0	–	2
CO	12	2	1	0
OT	50	0	3	–

Table 5: Development label substitutions for maximum-overlap pairs; no mismatch was predicted as ST or CO.

Use of AI tools. Generative AI tools assisted with early outlining, software development, and language editing, but supplied no annotations or hidden test references. The author made all experimental and system-selection decisions, ran or supervised experiments, verified every reported value against saved artifacts, edited the manuscript, and takes full responsibility.

8 Conclusion

BME-DALEEL combines joint paragraph/span prediction and constrained decoding with validation-gated specialists. Post-decode routing preserves the selected domain outputs exactly. The frozen system attained the reported official scores; future work should model overlap, short units, and rare-label count selection without test feedback.

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